focusing at large *NA* eliminates disturbances by aberrations or stimulated Brillouin scattering. Use of the bubble criterion for breakdown in conjunction with a sensitive measurement technique enables the detection of nanobubbles produced by non-luminescent plasma. Use of slm laser pulses eliminates statistical fluctuations of R_{max} and makes it possible to record smooth $R_{max}(E_L)$ curves. The shape of these curves can be understood by considering the interplay between multiphoton ionization and avalanche ionization with recombination and thermal ionization. Recently, Agrez et al. reported also separate thresholds for the formation of nanobubbles and shock wave emission plus large bubbles by tightly focused 515-nm, 60-ps pulses but without explaining the underlying dynamics of plasma formation [245].

The integral hydrodynamic response of optical breakdown reflected in the scaling law for cavitation bubble formation is similar for fs and ns plasmas but differs in the small-bubble region close to threshold. We shall see in the next two sections that the spatio-temporal characteristics of plasma formation itself, the resulting plasma substructure and the origin of plasma luminescence differ strongly also in the high-density plasma region.

5.3. Plasma structure, energy density and pressure in femtosecond breakdown

In fs breakdown, no plasma luminescence is observed near the bubble threshold. Only for energies $\geq 3\times E_{thB}$, we could photographically detect faint luminescence by integrating over many breakdown events (Fig. 7a). To the best of our knowledge, this is the first recording of plasma luminescence produced by nanojoule fs laser pulses focused into water. Previously, experiments were mostly performed with weakly or moderately focused laser beams [110,246] that are subject to an interplay of self-focusing, nonlinear absorption and plasma defocusing at super-threshold pulse energies. This interplay limits the free electron density in the relatively long plasma region and strongly reduces the intensity of plasma radiation [135,140,214,217,247]. Luminescence was reported only in one study, where water droplets were irradiated by a powerful collimated fs laser beam and back reflection at the droplet walls caused localized energy deposition within the droplets [222].

For laser-induced breakdown in bulk media, the plasma energy density is limited by the movement of the breakdown front during the laser pulse [108,132-136,248] (Fig. 13). This feature distinguishes it from breakdown at material surfaces during which a thin plasma skin layer evolves [6,137-139]. In the skin layer, high electron temperatures are produced, which results in X-ray emission [249-251]. During breakdown in bulk media by weakly focused ultrashort laser pulses, the breakdown front moves with the pulse because the plasma region is considerably longer than the geometrical length of the laser pulse

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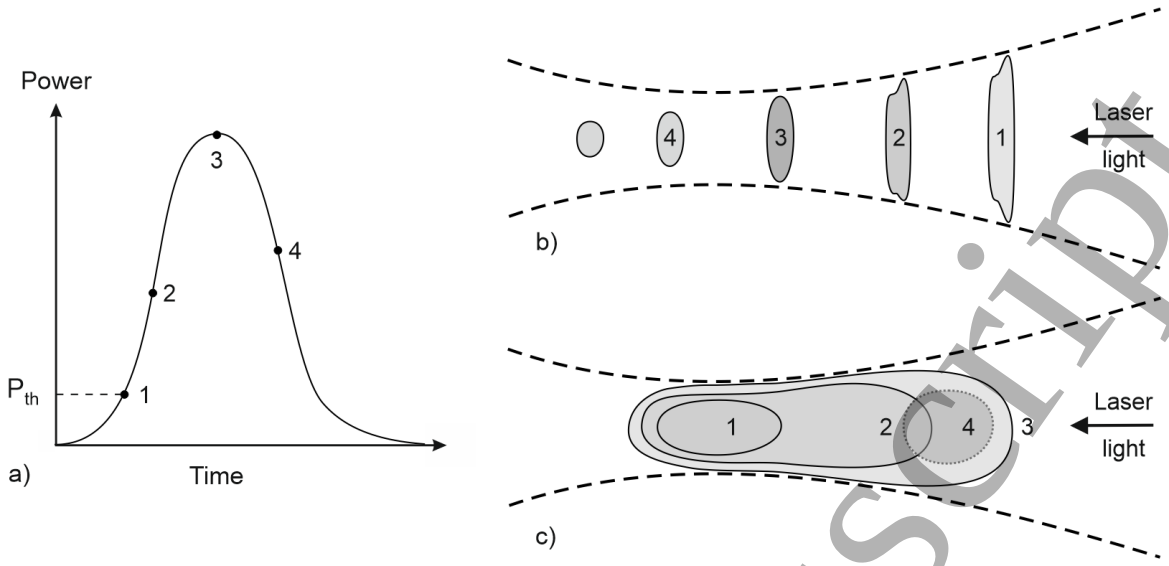

Figure 13 Optical breakdown dynamics in dependence on pulse duration and numerical aperture. a) Time evolution of the laser power during a superthreshold laser pulse. b) Dynamics for weakly focused ultrashort pulses with pulse length shorter than the plasma length. Plasma formation starts in the upstream region as soon as the irradiance of the focused pulse becomes large enough to exceed the breakdown threshold (1). While the pulse moves downstream and forms new plasma (2-3), its energy is ever-more depleted. At the same time, the ongoing focusing of the laser beam maintains a large irradiance of the travelling pulse such that plasma formation can continue up to the beam waist. In this process, linear focusing interacts with nonlinear focusing and with defocusing by the laser-induced plasma, which may result in filamentation (4). Nevertheless, electron and energy density are largest near the nonlinear focus. c) Breakdown dynamics for laser pulses with a spatial pulse length longer than the plasma length, which applies for long pulse durations and for fs pulses focused at large *NA*. Breakdown starts at the beam waist (1) and the plasma front moves upstream towards the incoming laser pulse (2), with the maximum extent reached at time 3. During the second half of the pulse, the irradiance is too small for further extending the breakdown region, and the laser energy is absorbed in the already existing plasma. This creates a hot spot in its upstream part (4), as seen in the plasma photographs in Fig. 8. Note that the drawings in (b) and (c) are not to scale; the actual focusing angle is much smaller and the plasma length much larger in (b) than in (c).

(Fig. 13b). As a consequence, the pulse is progressively attenuated by plasma formation while approaching the focus [135,248,252,253]. The details of this process are complex because linear focusing interacts with filamentation caused by the interplay of nonlinear self-focusing and plasma defocusing [214,218,252,254] and scattering by the plasma adds to the energy depletion [136]. With tight focusing, the plasma region is shorter than the geometrical laser pulse length. In this case, breakdown starts at the beam waist and the breakdown front moves upstream towards the incoming laser pulse [108,133,134] (Fig. 13c). This always happens in ns and ps breakdown but applies also for fs pulses that are focused at large *NA*. For focusing at moderate *NA*, when the spatial extent of a fs laser pulse resembles the plasma length, no clear upstream or downstream movement of a breakdown front is observed [135,136]. The

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"front" is blurred by the smooth onset of plasma formation in fs breakdown but the concept of a moving breakdown wave is very useful for identifying general trends in plasma dynamics. We will see in the following that upstream movement of the breakdown front produces much larger plasma energy densities than a downstream movement accompanied by nonlinear beam propagation and energy depletion.

For plasmas produced at NA = 0.8, the photographs of fs breakdown in Fig. 7a show no signs of nonlinear beam propagation and filamentation, and the plasma is only ≈ 5 µm long even for E_L = 500 nJ, twenty times above threshold. Hence, the plasma region is much shorter than the length of the laser pulse, which in water is ≈ 80 µm for a 350-fs pulse. Under these conditions, breakdown starts at the beam waist and the breakdown front moves towards the incoming laser pulse during the first half of the laser pulse, while the laser power increases (Fig 13 c) [102]. The velocity of the breakdown wave is fastest within the beam waist and slows down towards the far-field and towards the peak of the laser pulse. A quantitative description of the upstream movement of the breakdown wave has been given by Docchio et al. [133] and slightly modified by Vogel et al. to consider the finite length of the beam waist [108]. Both researchers derived relations for the dependence of plasma length on dimensionless laser pulse energy, $l_{plasma}(E_L/E_{th})$ at a given NA. These approaches portray the change of the location of iso-intensity lines characterizing the breakdown threshold during the laser pulse but do not consider nonlinear beam propagation, the lowering of the breakdown threshold by UV and X-ray radiation from plasma formed earlier during the pulse, and the change of direction of wave propagation in fs breakdown. The simple model yields reasonable agreement with experimental data for pulse durations of tens of picoseconds but are insufficient for fs breakdown at small and moderate NAs, where nonlinear beam propagation plays a role and for ns breakdown, where radiative energy transport comes into play [108]. For ultrashort laser pulses, spatiotemporal breakdown models are already available that describe the interplay of plasma formation with nonlinear beam propagation by Kerr-lens-induced self-focusing and plasma defocusing [102,130,131,136,141,156,252,255]. Models considering the influence of radiative energy transport during ns breakdown on plasma size are not yet available.

The average speed of the breakdown wave is given by the plasma length on the photographs divided by (τ_L/2). For the plasma formed at E_L =570 nJ, the average velocity of the breakdown front is 28570 km/s, which is 13% of the speed of light in water. The plasma growth ceases at the peak of the laser pulse, and during the second half of the pulse the light energy is absorbed in already existing plasma [108,133,134,188]. That results in a small plasma volume and high average energy densities between

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